AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-07-18

Abstract

Mission. A working end-to-end AWI-ESM3 (FESOM2–CORE3-cavity + OIFS-48r1 + LPJ-GUESS + WAM) ↔ Antarctic-PISM cycle: awiesm3 year → PISM decade → awiesm3 year on a changed mesh, with all three coastline-following members (FESOM cavity, OIFS land–sea mask, WAM sea points) regenerated per leg.

Where we are. Five real bugs have been found and fixed (§1); the wave-model coastline is solved structurally (§1.5); the workflow performs genuine awiesm3→PISM→awiesm3 handovers (§1.7); and the recurring OIFS crash family was traced past three partial fixes to its true root: **the ice–atmosphere surface-temperature coupling architecture itself is unstable in the marginal ice zone** (§1.6). The atmosphere's ice-tile skin was hard-pinned to the remapped FESOM ice temperature (skin conductivity 10^{10}), so kW-scale spurious fluxes circulated between the two models. The fix is architectural: OIFS now solves its own ice-tile skin implicitly each timestep, with slab conduction through the *coupled* FESOM ice thickness (new exchange field) and snow — the "coupled slab" (§2) — validated: the first slab run completed its year crash-free with the kilowatt flux events collapsed to zero.

The increment test now **passes end-to-end**: two stacked bugs — a layer-thickness mismatch that multiplicatively drains cavity tracers, and a bilinear mask remap that drowned Elephant Island — were found, fixed and validated (§4); the full cycle awiesm3 → PISM → awiesm3-on-changed-mesh closed and a 10-year/100-year run is underway. Open items: the *second* increment's Ross-front blowup (§4), and the too-weak Southern-Hemisphere summer sea-ice melt-back relative to the 500-year v3.4 reference (§5).

Contents

1 The campaign, in the order it happened

Each subsection carries the date it was worked. The bug numbering (I–V) is the order of *discovery* and is kept as the canonical naming.

1.1 The restart seam and the cold-start control (2026-07-08–13)

The original FESOM cavity-margin blowup: a remapped warm restart mixes an evolved, balanced dynamic state (98% of nodes, bit-exact) with patched values at changed nodes; the *seam* between them is the instability. The decisive control:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up restart	stable, worst CFLz 2.61
PISM cavity, cold	**1200 s**	**cold start**	**stable, CFLz 3.44**
PISM cavity, warm	1200 s	remapped restart	dies day 4.0 (CFLz 9.2)
PISM cavity, warm	60 s	remapped restart	dies day 4.4 (η_n, 0 CFLz)

The ocean lives happily in PISM's cavity; the production answer became: run chunk-1 cold-started on the PISM cavity (artifacts pooled), and let `remap_restart` carry only genuine per-leg increments. The first real increment test crashed on 17 fully-opened columns (ice removed entirely, 16 levels at once) out of 1399 — but its numbers are TAINTED (measured under Bugs I+II) and the finding that matters survived into the fixes below: absorbing 99% of a geometry change is not a passing grade when the remaining 1% is fatal.

1.2 Bug I — PISM was fed surface water as ice-base forcing (2026-07-14)

`coupling_ocean2pism` built PISM's `-ocean th` forcing as a mean over FESOM level *indices* 1–20 = **0–410 m including the summer surface** (the correct `-sellevel,150/600` sat in dead code). PISM received ice-base water up to +6.5°C even from a healthy ocean, melted at ∼30 m/yr, and shed **24% of its floating area per decade** — every mesh-change leg then had to absorb a catastrophic geometry increment. FESOM's own cavity melt in the same runs was sane (2.2 m/yr mean, 0.72 median).

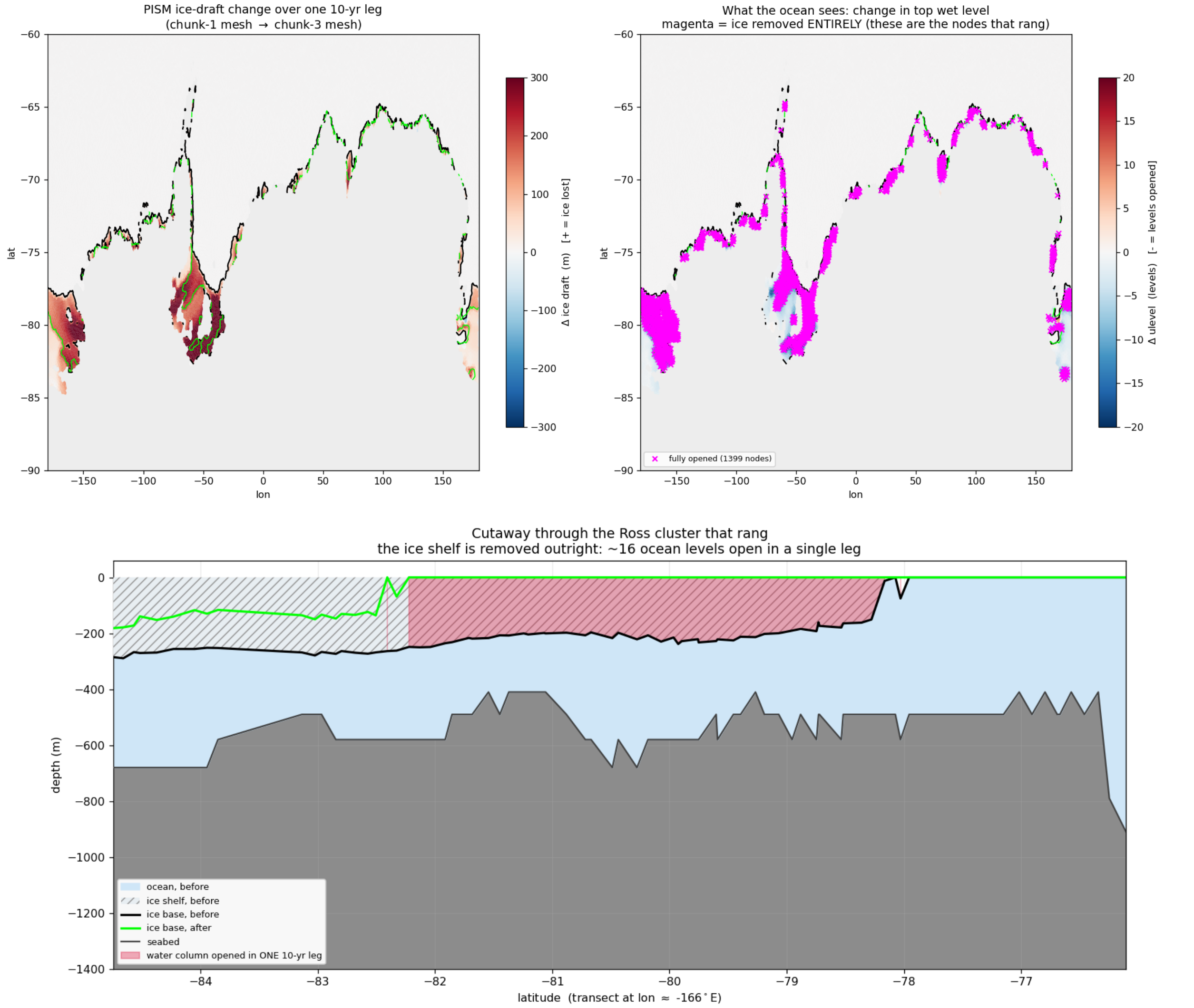

Figure 1: What Bug I did to PISM in one 10-year leg under surface-water forcing: 1399 nodes lose their ice entirely (magenta ring around the whole margin), the Ross front retreats ≈440 km (section: ice base $-200\ldots-280$ m → open ocean across four degrees of latitude).

Fix (`93a207ee`): the DIRECT route. PISM consumes FESOM's own cavity fluxes via `-ocean given`: `fw`→`shelfbmassflux` (×1000, sign preserved), `sst`→`shelfbtemp`. One model owns the ice–ocean interface. The `-ocean given` machinery existed all along but silently fell back to `th` because its input is a FESOM1.x-era bundle FESOM2 never wrote; `fesom2ice` now synthesizes it (plus three exhumations of never-run 2D-path code: `3f07d7a2`, `74671c27`, `bb9d00c2`).

1.3 Bug II — the exchange weights scrambled the coupled ocean (2026-07-14)

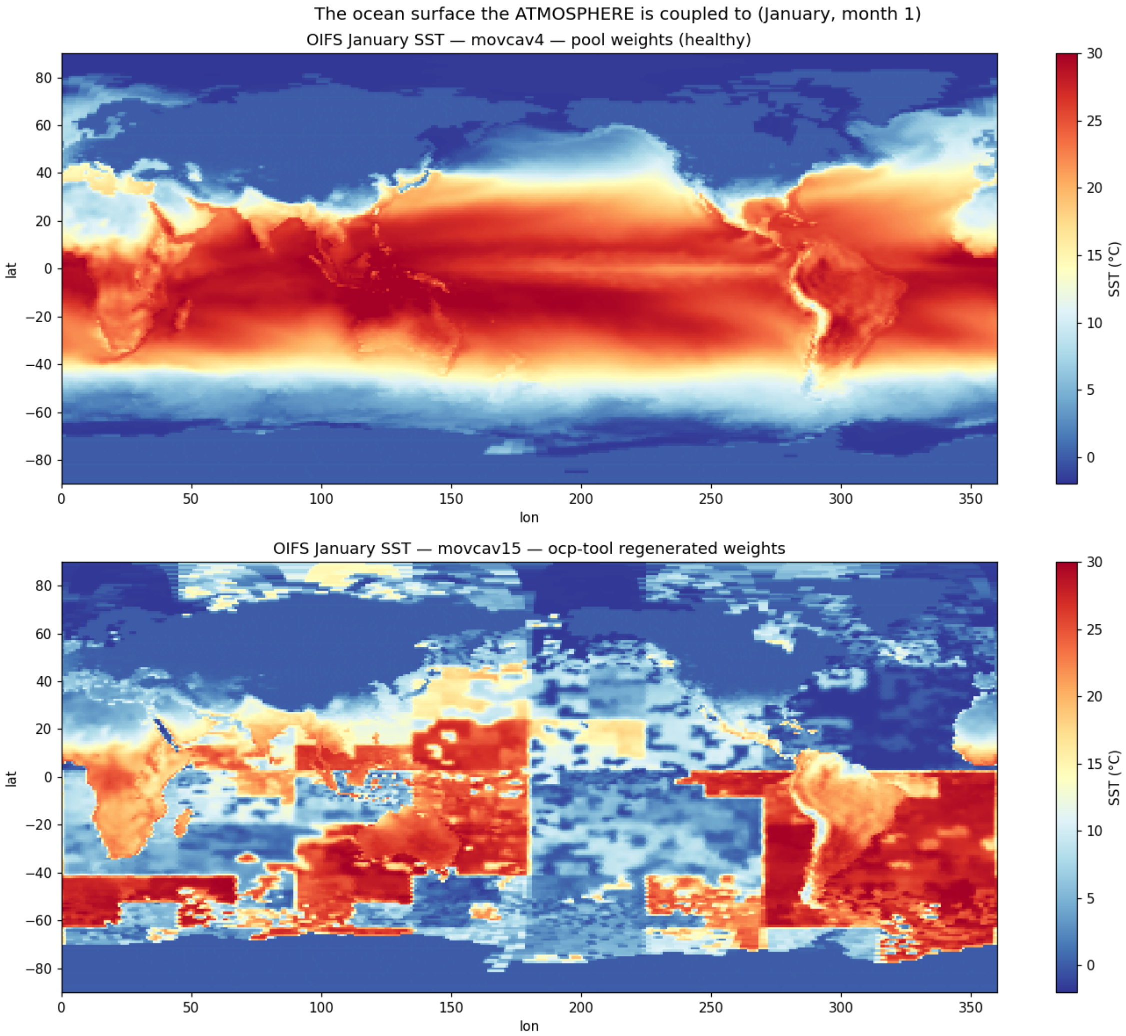

Figure 2: **The picture that settles it.** January SST as OIFS holds it. Top: pool weights — textbook. Bottom: offline-engine weights — a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a seam at lon 180 where the node ordering breaks.

FESOM exchanges coupling fields in *partition (my_ list/rank) order*; the offline weight engine (2026-06-17) built the feom geometry in *nod2d.out order* (median displacement 58°). Every submesh exchange was geographically tile-translocated (Fig. 2): OIFS saw Weddell ice fraction 0.04 while FESOM had 0.95; global sea-ice death followed within months.

Fix (`c21fe7c2`): `permute_feom_to_runtime.py` reorders the OASIS geometry to runtime order (via `dist_<nproc>/rpart.out`); `validate_feom_weights.py` applies every weight file to real latitudes and **aborts the leg** above 1° median error (nod2d-ordered weights fail at 42–58°, correct ones pass at $\leq 0.025°$).

1.4 Bug III — the OASIS coupling restart was compound-scrambled (2026-07-14)

`remap_oasis_restart.py` gathered in nod2d space, but its input and required output are both *run-time*-ordered: the emitted `rstos.nc` matched *neither* ordering (correlation of `sst_feom` with the restart's own surface temperature: +0.15). OIFS's first coupling interval saw 305 K water under polar air — a one-hour global jigsaw shock that nondeterministically detonated the convection scheme (identical inputs crashed at hour 19, hour 458, or never).

Fix (`abe56ee2`): un-permute via the old mesh's `rpart.out` → gather → re-permute via the new mesh's; correlation now +0.995. Pool `rstos.nc` rebuilt.

1.5 The wave model's frozen coastline (2026-07-15)

Chunk-3 died deterministically at its first physics step in `voskin_mod.F90:282` squaring WAM's Stokes drift. WAM has no runtime tiling: its sea-point set is baked into `wam_grid_tables` at preprocessing time and had never been regenerated. A cell that is ocean per the regenerated lsm but land per WAM never receives wave data; the Stokes/Charnock fields are srf-restart-carried, so a *warm* start after a coastline change reads never-written storage (cold starts zero-initialize — why chunk-1 tolerates its 12 mismatch cells and historical lsm changes never hit this). Chunk-3 had 125 mismatch cells. WAM itself was separately exonerated for the Bug-IV crash family (a WAM-off run crashed identically).

Fix: one-time maximal-ocean tables (option A; tolerance patching rejected). WAM's sea-point set was rebuilt from ETOPO1 with the southern limit extended 78 → 85°S, then *max-ocean edited*: every cell the FESOM max-mesh can make wet — open ocean or sub-shelf cavity — is forced to sea at its FESOM bottom depth (only ever adding ocean). 1277 cells opened (29 555 → 30 832 sea points), 283 of them south of 55°S: exactly the band a moving coastline walks through (Fig. 3). The tables never change again, wave restarts stay index-compatible across all future mesh changes, and WAM cold-starts once at the switch leg. Known cosmetic debt: the max-ocean edit did not update the obstruction coefficients.

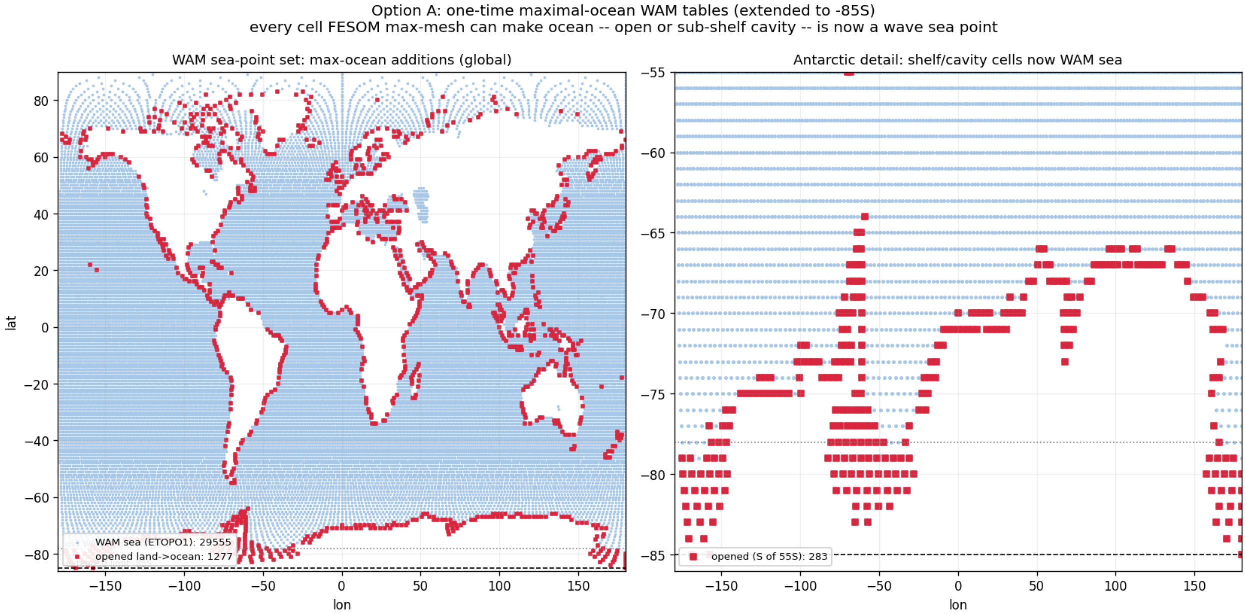

Figure 3: One-time maximal-ocean WAM tables (extended to 85°S). Blue: wave sea points from ETOPO1. Red: cells opened land→ocean because the FESOM max-mesh has a water column there. Pre-registering them makes the `voskin` step-0 crash structurally impossible on any future leg.

1.6 Bug IV — the ice-surface-temperature coupling is architecturally unstable (2026-07-14→16)

This is the mid-run OIFS crash family (`cubasen`/`vsurf`/`voskin` floating overflow, random timing, polar/MIZ ranks). It survived three partial fixes before the root was found; the full chain is worth recording because each layer was real, just not sufficient.

Layer 1: MIZ aliasing of the sent temperature. FESOM sent raw `ist`; ice-free nodes send the seawater freezing point, so the GAUSWGT cell blend in the marginal ice zone was dominated by warm open water: OIFS's ice tile saw a surface tens of K too warm and returned a strongly negative flux to the fully-iced nodes. Partial fix: concentration-weighted ist (`ist`·`a_ice` ÷ ci on ingest, the EC-Earth/NEMO convention behind `ECE_CPL_NEMO_WEIGHTED_ICE`). Runs survived 3× longer, still crashed.

Layer 2: the FESIM solve has no flux feedback within the coupling interval. FESIM's skin solve relaxes toward $t^* = T_f + \mathrm{a2ihf} \cdot \mathrm{zsniced/con}$ with the received flux *frozen* for the hour and no $\mathrm{d}Q/\mathrm{d}T$ term: under insulation the equilibrium runs away (measured floors 145–183 K; excursions to −954 K in one run). Partial fix: an implicit flux linearization anchored at the transmitted temperature, $Q(t) = \mathrm{a2ihf} + \lambda\,(t_{\mathrm{ref}} - t)$ with $\lambda = 4\varepsilon\sigma t_{\mathrm{ref}}^3 + \lambda_{\mathrm{turb}} \approx 20\,\mathrm{W/m^2/K}$ (the NEMO-family fallback derivative), energy-closed by booking the linearization heat to the ice growth budget (fesom branch `awiesm3-implicit-ice-surftemp`, `38a558b5`). Dips became bounded and self-recovering in most runs — but not all.

Layer 3 (the root): the atmosphere's ice skin was never solved at all. Send-side instrumentation of `A_Q_ice` (term decomposition + tile state at every $|Q| > 1\,\mathrm{kW/m^2}$ event) delivered the verdict:

- The received flux at cold FESOM nodes is **unphysical and anti-braking**: mean $-15.8\,\mathrm{kW/m^2}$ below 150 K (100% negative), −4.5 kW at 150–180 K, recurring all year with the MIZ seasonal cycle — at **49°N on 0.1–0.5 m ice with thin snow** (Okhotsk / St. Lawrence). Insulation is irrelevant there; no realistic λ brakes kilowatts.
- At the OIFS send, the generating cells have ice-tile fractions down to 0.04 and **tile skin temperatures of ∼204 K under ∼275 K maritime air**; the sensible + latent terms then reach $+17.8\,\mathrm{kW/m^2}$ per tile area. With weighted-ist the tile skins were *still* 189–204 K at substantial tiles — the cold temperature is genuine FESIM state, faithfully communicated.
- The reason the skin can sit 80 K below the air: `vlamsk_mod.F90`, sea-ice tile, with `LNEMOSICOUP=.true.` (the default): `PLAMSK = 1e10` — **the skin conductivity is set to infinity, pinning the tile skin to the prescribed (remapped FESOM) ice temperature**. The atmosphere's surface scheme was forbidden from solving its ice surface; it computed fluxes against whatever arrived through the remap.

The loop, in one line: FESIM's oscillating solve → remap → *pinned* OIFS skin → kW fluxes against an unresolvable surface → remap → FESIM slams further — a two-model positive feedback closed through the marginal ice zone, metastable enough that identical configurations coin-flipped between completing a year and dying at day 2 (runs movcav25–31: raw/weighted × $\lambda = 3.4/20$ × dirty/clean initial state — every combination crash-prone).

Root fix: the coupled slab (§2). Stop prescribing the atmosphere's ice skin altogether; let OIFS solve it implicitly each timestep with conduction through the *coupled* FESOM ice and snow thickness. This removes the loop by construction while keeping the flux honest about the actual ice state. Upstream note: mainline AWI-CM3 carries the same pinned-skin architecture and therefore the same latent (if rarer) crash mode.

1.7 Bug V — iterative coupling never handed over to PISM (2026-07-16)

The awiesm3→PISM model switch is resolved at SimulationSetup init by reading the `chunk_date` file, whose path `esm_runscripts/chunky_parts.py` builds by *raw string concatenation of* `general.base_dir` *before esm_parser resolves variables.* A base_dir containing `${user}` was probed verbatim; the silent `isfile()` miss fell back to "chunk 1, model 1" — awiesm3 self-resubmitted to its own final date and PISM never ran. Proven by log comparison: identical resubmission command lines resolve to `Valid Model Names = ['pism']` under a literal base_dir (movcav16 era) and to `['fesom','oifs',...]` under `${user}`.

Fix (`d61297e8`): `_early_resolve()` expands `${...}` against `general` (environment fallback) in all three chunk-file path builders. Regression-tested against a real chunk file, and validated live: the first genuine awiesm3→PISM handover of the campaign (movcav27), and a full awiesm3→PISM→couple_out→chunk-3-couple_in chain (movcav29).

2 The coupled-slab architecture (2026-07-16/17)

The new division of labour: **OIFS owns the ice surface temperature; FESOM owns the ice state.** OIFS's surface scheme solves the ice-tile skin implicitly within every atmosphere timestep (full radiative + turbulent $\mathrm{d}Q/\mathrm{d}T$, finite skin conductivity), with slab conduction through the coupled per-ice thickness and snow depth. FESOM's albedo still drives the radiation; FESIM's own ist solve remains for its internal state (albedo, melt ponds) but is read by nobody — the feedback loop is cut by construction. The machinery is EC-Earth/KNMI heritage (`ICESTATENEMO`, 2008; the SI3 branch of `callpar`) — our contribution is wiring the FESOM branch, which lacked only the thickness.

New exchange field. FESOM sends `sit_feom` = effective (grid-mean) ice thickness `m_ice` (o2a collection, `A_Ice_thickness`); `ECE_FESIM_GET_ICE_STATE` divides by the received ice fraction (standard ε-guard) for the per-ice thickness the slab needs, falling back to the climatological 1.5 m only where the cell has no significant ice. Snow thickness (`snt_feom`) was already exchanged and now actually reaches `surfseb` via `SURFL%ZSNTICE`.

The switch set (verified in code, not inferred):

switch	setting	verified effect
`LNEMOSICOUP`	**.false.**	`vlamsk`: sea-ice tile skin conductivity becomes *finite* (land-tile-style lookup) instead of 10^{10} — the skin is genuinely solved by the SEB. The prior default pinned it to the prescribed ice temperature; this pin was the kilowatt engine of Bug IV.
`LNEMOLIMTEMP`	**.false.**	disables the hourly overwrite of OIFS's 4-layer ice temperature (`SP_SB YTL`) from the FESOM ist ingest; the profile is OIFS-prognostic (ICMGG-initialized, restart-carried). FESOM's ist send becomes inert.
`LNEMOLIMTHK`	**.true.**	activates the `callpar` fill of `ZTHKICE/ZSNTICE` from the coupled fields — the skin/top-layer conduction feels the real ice thickness and snow insulation. (The deep 4-layer discretization keeps its fixed depths; the skin conduction is the term that controls winter fluxes.)
`LMCCDYNSEAICE`	.true.	master enable; false force-resets the whole `LNEMOLIM*` family (`sumcc`).
`LNEMOLIMALB`	.true. (default)	FESOM's albedo drives OIFS radiation (`surfrad_layer` → FESIM getter; raw–raw convention, consistent).
`ECE_CPL_NEMO_WEIGHTED_ICE`	.true. (inert)	only acted inside the disabled ist ingest.
FESOM `latm_owns_ist`	**.true.**	FESIM growth budget takes the atmosphere flux directly (`Qatmice=-a2ihf`, the ECHAM convention); qres/qcon/qlam booking off.

The `sumcc` guard "`LNEMOSICOUP` without `LNEMOLIMTEMP` does not make sense" correctly enforces that a pinned skin requires a prescribed temperature; our mode flips both off. In one sentence: *before, OIFS's ice surface was a mirror bolted to whatever FESOM's oscillating solve sent; now it is a real solved surface conducting through FESOM's actual ice and snow.*

Validated (movcav33). The first coupled-slab run completed its year crash-free and handed over to PISM. With the instrumentation still armed: kW-scale send events at real ice tiles went from 1410–1930 per run to **zero** (117 residual events are all phantom slivers with ice fraction $\sim 10^{-7}$, a `ZMASK` threshold cosmetic); event tile skins moved from 189–212 K into the physical 247–265 K range (Fig. 4); the fluxes FESOM receives at its coldest nodes are $-63 \ldots -80\,\mathrm{W/m^2}$ (textbook polar values, was $-10\,\mathrm{kW}$), and FESIM's internal temperature tracks its anchor to $< 0.1\,\mathrm{K}$.

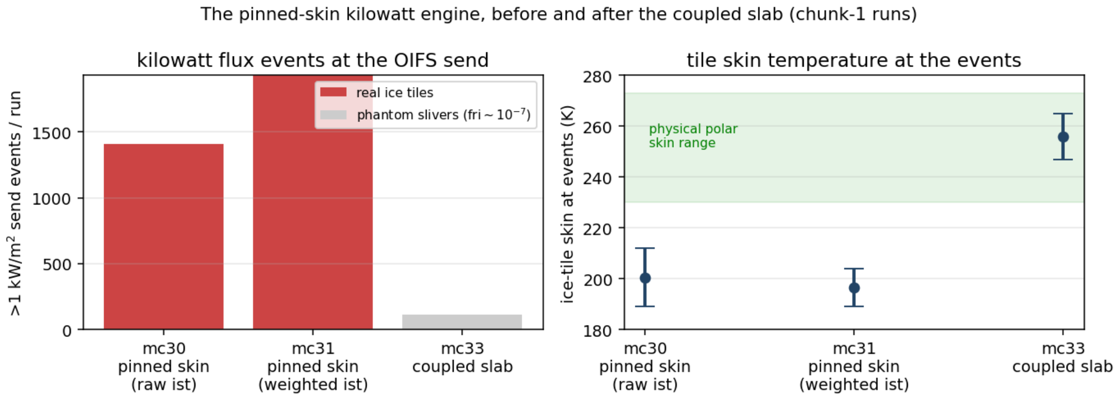

Figure 4: The kilowatt engine, before and after. Left: $>1\,\mathrm{kW/m^2}$ events at the OIFS send per chunk-1 run. Right: the ice-tile skin temperature at those events — pinned-skin runs sit 60–80 K below the physical range; the solved skin cannot.

3 First fruits of the clean decade: cavity melt against observations (2026-07-17)

The chunk-1 year that forced the first clean PISM decade, split by sector and set against the satellite record (Rignot et al. 2013; Adusumilli et al. 2020):

sector	observed	movcav33 chunk-1 (m w.e./yr)		
	(m ice/yr)	mean	median	nodes
Filchner–Ronne	0.3–0.4	0.89	**0.20**	1085
Ross	∼0.1	0.90	**0.31**	1075
Amundsen–Bellingshausen	14–27 (PIG 16)	7.8	6.4	229
Antarctic Peninsula	0.4–3.8	7.5	4.4	186
Wilkes (Totten sector)	4.7–9.1	8.3	6.4	296
Amery	0.6–0.7	4.0	3.3	128
E. Weddell / Dronning Maud	0.3–0.9	4.4	3.5	287
circum-Antarctic	0.8 ± 0.7	2.84	0.79	3286

The cold/warm dichotomy that dominates the observations is reproduced: the cold-cavity giants sit at 0.2–0.3 m/yr medians while the warm sectors run 4–8, and 13% of the cavity area refreezes (marine ice, as observed under Ronne/Amery). Residual biases: the East Antarctic coastal sectors (Dronning Maud, Amery) run ∼5× warm, and the Amundsen runs ∼2× cold — both worth revisiting once multi-year coupled statistics exist. Caveats: node statistics (unweighted), m w.e. vs the observations' m ice (×0.92), one coupled year, and PISM's initial geometry carries reduced versions of the big shelves. For scale: the jigsaw-era coupling produced 28 m/yr.

4 The increment test crashes, twice — and passes (2026-07-17/18)

The first chunk-3 leg — awiesm3 on the PISM-changed 208 438-node mesh, stage 6/6 — absorbs the mesh change and runs for weeks of model time before FESOM dies of an SSH explosion (η jumping ±40 m in one step at the Antarctic Peninsula tip; day 19–51 depending on MPI reduction order — the state is metastable). The autopsy was cheap for two reasons: the leg costs 3–4 minutes of wall time, and a temporary XIOS file group with `sync_freq="1d"` flushes daily fields to disk as they complete, so even a crashed run leaves full evidence (default XIOS files sync only on close: 51 days of output, zero records). The hunt found *two independent bugs*, stacked at the same place.

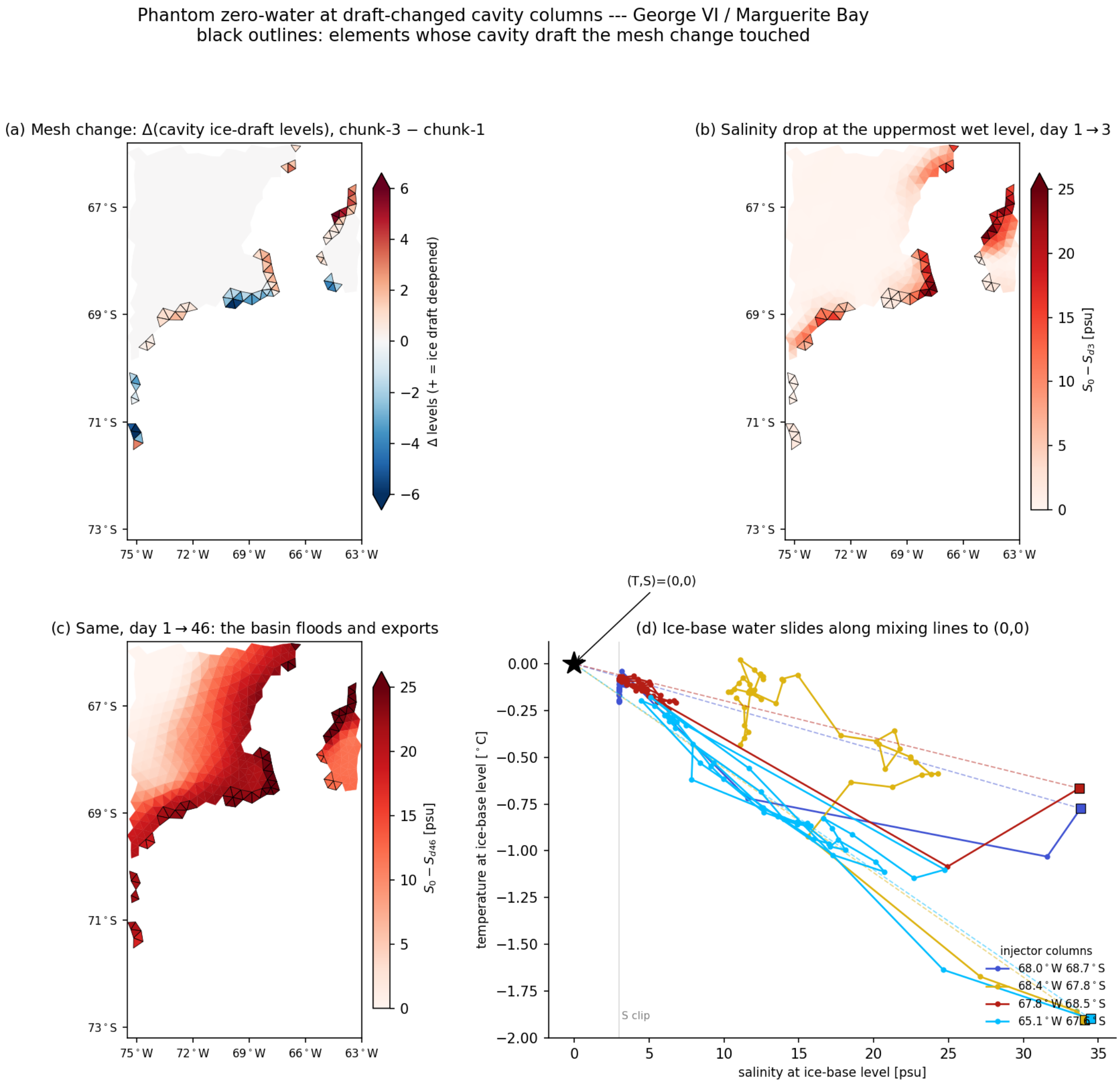

Figure 5: Bug VI seen in one basin (George VI / Marguerite Bay; black outlines = the 73 elements whose cavity draft the mesh change touched). (a) The mesh change: ±1–6 cavity levels. (b) By day 3 the uppermost-wet-level salinity has dropped up to 25 psu *exactly on the outlined elements.* (c) By day 46 the basin has flooded and exports along the coast. (d) The fingerprint: (S, T) at the ice-base level slides along the mixing lines toward $(0, 0)$ with *identical* T- and S-replacement fractions — not a flux (surface freshwater is 50× too small and negative), but a *multiplicative drain*: both tracers lose the same fraction of their value every step.

Bug VI — the layer-thickness mismatch drain

The remapped restart carries `hnode` inherited from the old mesh's zstar state (e.g. 9.964 m where a formerly ice-free column scaled its layers with η). At runtime, cavity columns and their open-ocean edge ring (`ulevels_nod2D_max>1`) keep layer thicknesses *fixed at the mesh nominal* (10.000 m) and are excluded from the per-step `hnode`↔`hnode_new` sync — so the mismatch is immortal. FESOM's tracer T* update applies values · (`hnode` − `hnode_new`)/`hnode_new` every step: a −0.36%/step multiplicative drain on T *and* S. Per-operator probes at a Getz injector column measured −0.1226 psu/step against a predicted 33.64 × 0.0036 = −0.1224; an unchanged FRIS control column measured exactly zero.

At free-surface nodes next to the cavity edge the drain even *accelerates*: η grows, the nominal-vs-restart gap grows with it. **Fix** (FESOM, `restart_thickness_ale`): after the restart read, force `hnode` at the whole excluded class back to the fixed nominal and rebuild the depth arrays. Validated by intervention: the drain vanished (+0.0002/step), day-1 fields healthy.

Bug VII — the drowned island

With Bug VI fixed the run survived longer but still exploded at the Peninsula tip, η climbing a steady +0.11 m/day. The reason is geometric, not numeric (Fig. 6): the ice2fesom regrid remapped PISM's *categorical* mask **bilinearly** — a few-cell island surrounded by mask-4 ocean smears to ≈3.x, rounds to ocean, and is deleted from the submesh. PISM's own bed there is +38 to +218 m above sea level: bedrock island, regardless of ice. (The island entered the campaign *bare* — the spinup state carries 0.0 m of ice on all ten cells; a thin 0.1–0.4 m cap has been *growing* during the coupled decades. A second, independent deletion path was found while examining it: PISM encodes ice-free bedrock as mask=0, which the node classifier silently defaulted to ocean; fixed — bare bedrock above sea level is land.) The ocean then drives a runaway coastal jet where a coastline should be. **Fix** (ice2fesom): remap the categorical mask nearest-neighbour; rebuild the mask=5 outside-domain marker (which `remapnn` extrapolation would erase) from a bilinear domain indicator. Continuous fields stay bilinear.

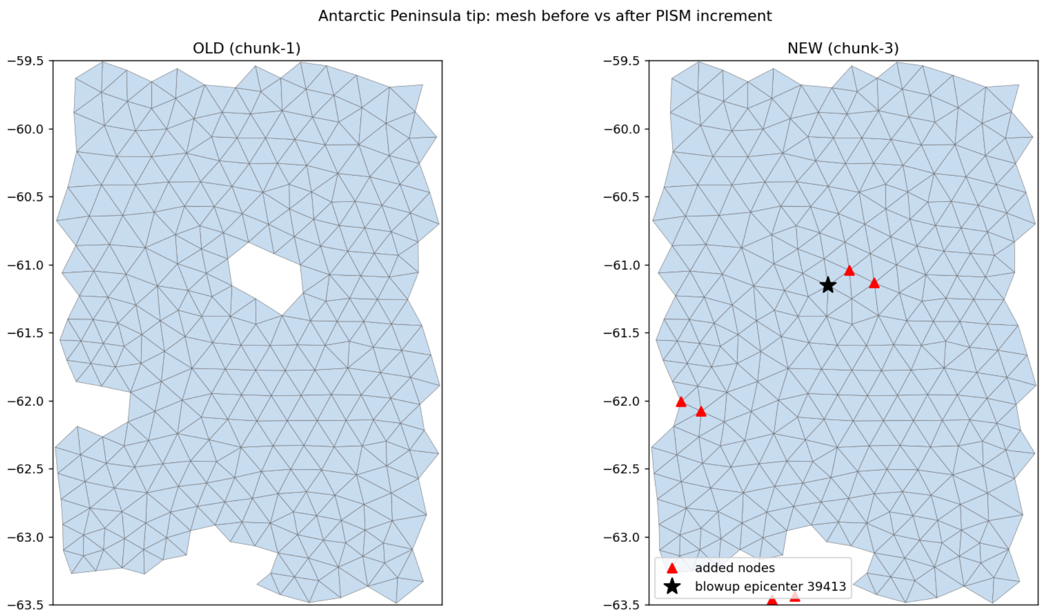

Figure 6: Bug VII. Left: the chunk-1 mesh has Elephant Island (the hole at 55.4°W, 61.1°S) and the Clarence/Gibbs notch. Right: after the PISM decade the bilinear mask remap has rounded both islands to ocean — the added nodes (red) fill them in, and the blowup epicenter (star) sits exactly on the drowned island.

Ruled out by measurement (each was a plausible suspect): surface freshwater fluxes; OIFS runoff/calving and the `ECE_LANDICE` snow cap; the OASIS `feom` masks; the remapped T/S state (freezing cap ran: 1137 columns); mesh-file level-range consistency (zero violations); the zero-filled dry restart levels (padding them — kept as a permanent hygiene pass — reproduced the crash bitwise); and the vertical-advection top boundary (cavity guards in both the implicit and explicit schemes changed nothing — see §7). The couple_in warning about a missing `cap_new_cavity_temp.py` is a red herring: the freezing cap runs as Fortran inside the remap binary.

The increment test passes

With both fixes, movcav33 chunk-3 completed the full year 1901 on the changed mesh (25 min wall, no blowup; the Peninsula node's SSH wanders seasonally instead of pumping) and handed over to the chunk-4 PISM decade unattended — **stage 6/6 closed**. The long run movcav34 (10 awiesm3 years / 100 PISM years) then repeated chunks 1–4 cleanly on the *minimal* build (hnode fix only; the advection guards were reverted after being falsified). Fixes are committed: FESOM `awiesm3-implicit-ice-surftemp`, esm_tools `development_interactiv_mesh_awiesm3`.

New open item: the second increment (Ross front)

movcav34's chunk-5 (year 1902, after 20 PISM years) blew up at day 16 at the western Ross front, where the second increment opened cavity columns wholesale (ulev 13 → 1) amid a 506-node ice advance — the new coastline forms a pinched notch at the epicenter (Fig. 7). This is the next campaign front: candidate responses are a coastline-hygiene pass in the submesh generation (forbid single-node notches and pinches) and moderating PISM's per-decade geometry increments.

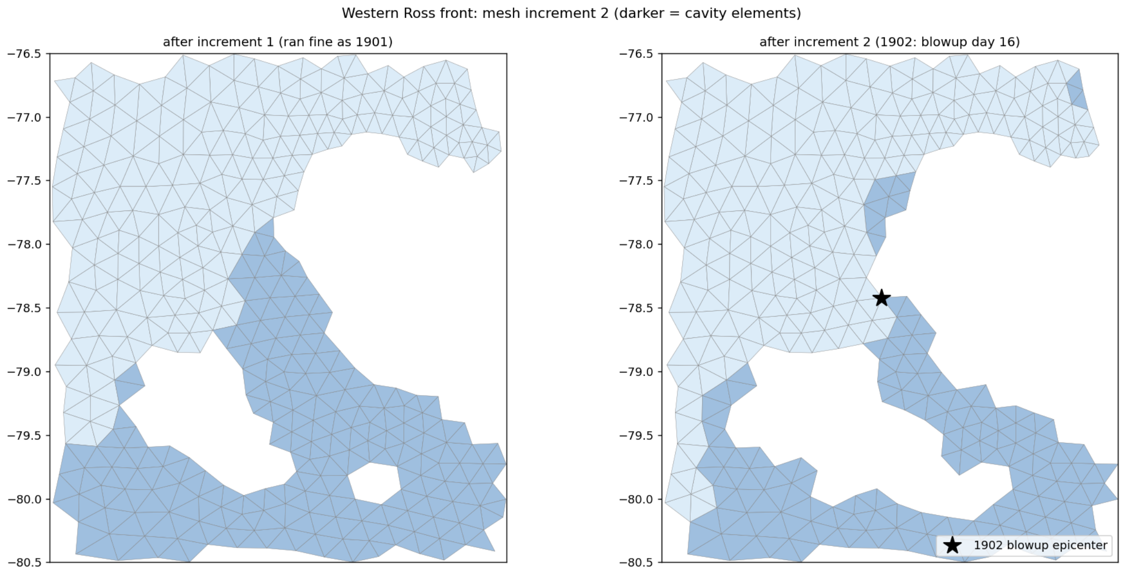

Figure 7: The second increment at the western Ross front (darker = cavity elements). Left: after increment 1 (ran a clean year as 1901). Right: after increment 2 — the front reorganizes, and the 1902 blowup epicenter (star) sits in a newly pinched notch between advancing and retreating ice.

5 State of the cycle and of the climate (2026-07-18)

5.1 What is proven

- **Chunk-1 is quantitatively healthy and, under the coupled slab, crash-free**: January SST physical, zero translocated cells; OIFS Weddell September ci 0.61/0.99 (was 0.04); both-hemisphere sea-ice annual cycles; per-basin cavity melt in the observed regimes (§3); kilowatt flux events at real ice tiles: zero.
- **The workflow hands over in both directions** (Bug V fixed): awiesm3 → PISM → couple_out → chunk-3 couple_in runs unattended.
- **The PISM decade runs on FESOM's own melt** (`-ocean given`, 2026-07-17): real output,

and a *gentle* geometry response — 290 fully-opened nodes vs 1399 in the Bug-I era, mean $|\Delta \text{draft}|$ 1.0 m, submesh 208 810 → 208 438 nodes (Fig. pism_change_mc33).

- **The live mesh change works**: weights regenerated and runtime-order-validated ($\leq 0.024°$), restarts remapped through the rpart sandwich, ICMGG/plit regenerated.
- **The increment test passes end-to-end** (2026-07-17/18): chunk-3 runs its full year on the changed mesh and hands to chunk-4 PISM unattended; movcav34 (10 awiesm3 years / 100 PISM years) repeats chunks 1–4 cleanly on the minimal committed build. The second increment's Ross-front blowup (§4) is the current front.
- ~~The chunk-3 dilution enters through the vertical-advection top boundary (zero-tracer inflow ratchet / material-interface flux).~~ Cavity guards in both the implicit (`adv_tra_vert_impl`) and explicit (`upw1`, `qr4c`) schemes changed the crash bitwise-nothing; the guards were reverted. The identical-fraction $(T, S) \to (0, 0)$ signature belongs to the *multiplicative* `hnode` mismatch drain (Bug VI).
- ~~The dilution is unclosed column continuity: remnant $\nabla \cdot (u\,h)$ at pinned-η cavity columns drains tracer content at fixed volume.~~ The divergence and ice-base w are symptoms of the same `hnode` inconsistency, not an independent leak; with the restart `hnode` reset the columns are quiet.

5.2 Open item: the Southern-Hemisphere summer melt-back

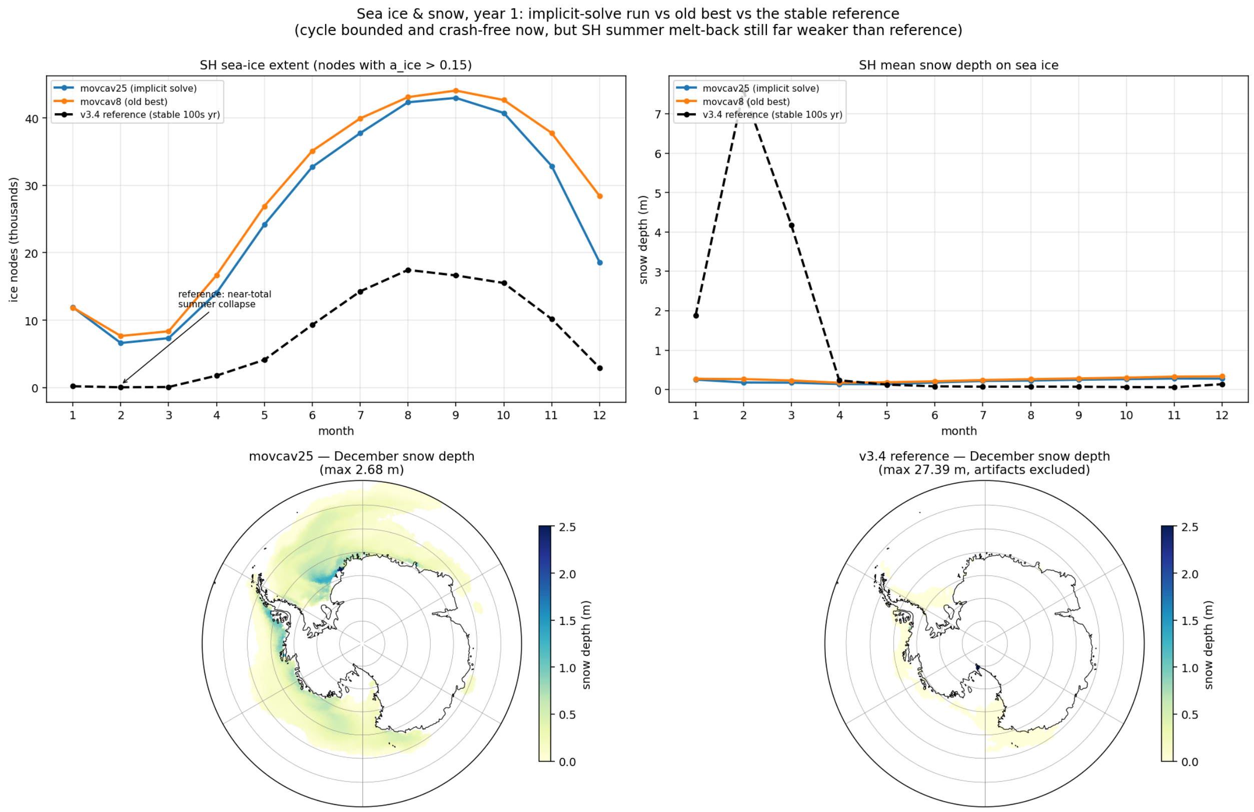

Figure 8: Year-1 sea ice and snow: implicit-solve run (blue) vs the movcav16-era state (orange) vs the 500-year-stable v3.4 reference (black dashed). Our runs track each other — the crash fixes did not degrade climate — but the reference collapses its SH ice almost completely each summer while ours retains ∼7k nodes, and its winter pack carries 0.03–0.08 m of snow against our 0.2–0.3 m. Bottom maps (same scale): December snow depth — our thick band hugs the Antarctic coast; the reference is nearly bare.

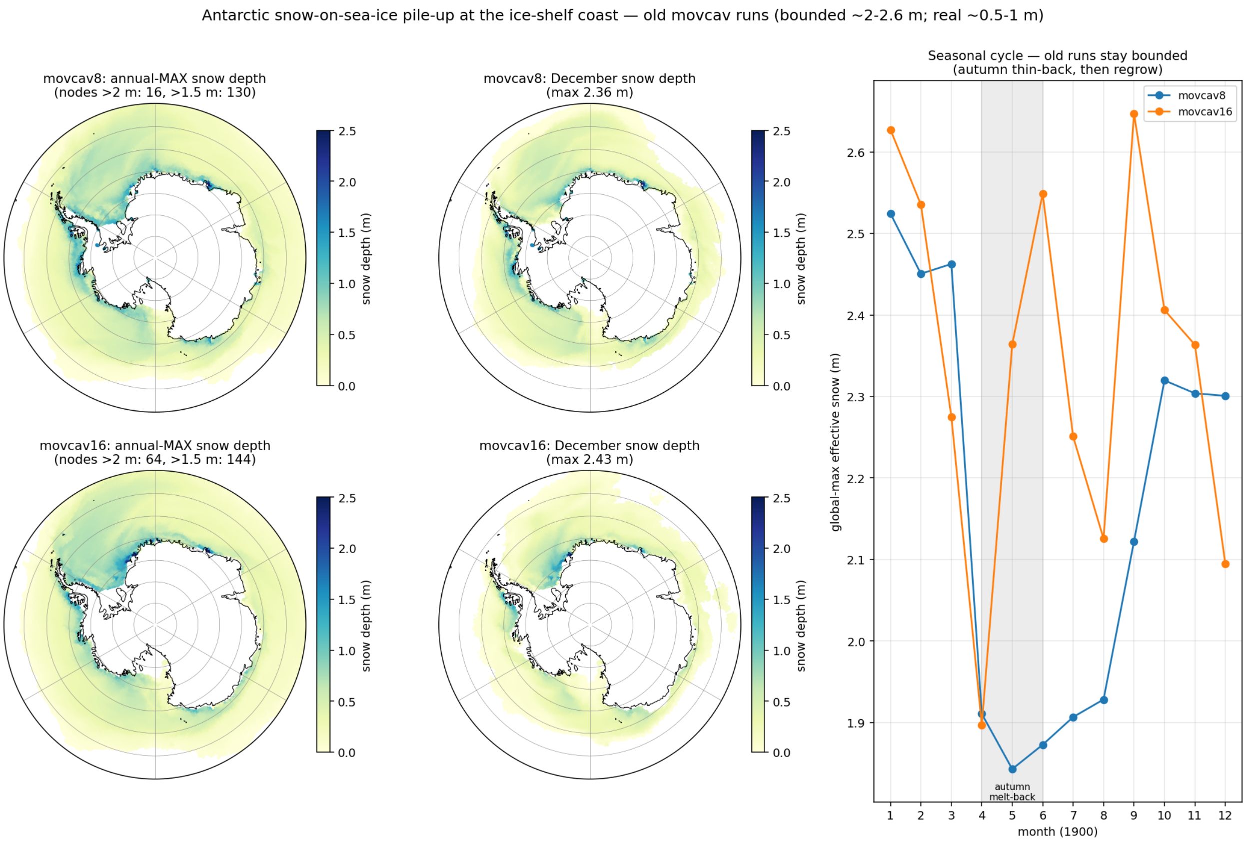

Figure 9: The extent of the coastal pile-up in the (pre-fix) movcav runs: annual-max and December snow depth, and the seasonal cycle. The thick snow is a band hugging the ice-shelf front (eastern Weddell/Fimbul, Amundsen), bounded at 2–2.6 m by an autumn melt-back. Later runs that lost this melt-back drifted monotonically to 3.5 m — the insulation that raised Bug IV's loop gain.

The ice-thermodynamics code is byte-identical to the v3.4 reference; the difference is state and forcing. Ours keeps ∼2.5× the reference's winter SH extent and only a ∼6× summer retreat where the reference manages ∼200× (Fig. 8), and piles 2–3.5 m of snow against the Antarctic ice-shelf coast (Fig. 9). Two mitigations exist: a *clean chunk-1 ice initial state* (reference year-1849 fields index-mapped onto the submesh via `map_nod.out`, 7 Ronne artifact nodes hybridized; snow max 0.81 m — in use since movcav29), and the coupled slab itself (the pinned-skin era's fictitious equilibria drove a spurious congelation-growth term). Whether these restore the reference's melt-back is the first climate question for the post-slab runs; meltponds were checked and are *not* the SH lever (the `hsno>hs1` lockout makes old and corrected schemes identical there).

6 Secondary bugs, fixed along the way

Bug	Fix	Commit
Remap extrapolated unboundedly below the old seabed (manufactured −45°C) when columns deepen	take water from the nearest old node wet at that depth	`eef4f6db`
Sub-ice η/`hbar` kept open-ocean values at newly-sub-ice nodes	zeroed (real bug, not the cure)	`b17fb5c8`

Bug	Fix	Commit
First mesh-change leg read the mother mesh as cavity-free (`cavity_nlvls.out` missing from the `mother_as_old` links) ⇒ zeros in newly-wet levels, mstep-1 blowup	link the cavity files	functions
Same-chunk `couple_in` rerun promoted its own submesh to `previous_submesh` (old = new ⇒ zeros again)	rerun guard	`b9c1e370`
OASIS `GSSPOS` conservation blowup on near-zero target residuals	`gauswgt_gsmart` transform	`3e94d4da`
namcouple `feom` dimension never actually patched (env var from another subjob's shell)	fixed variable passing	`3cf29e9e`
`fesom2ice` loaded the static full mesh against submesh output	load the run mesh	`63cfcc91`
`previous_submesh` read but never created	created; plus restart size guard	`3b44a704`
Failed coupling functions did not stop the workflow (backgrounded, observe saw no exit code)	`.coupling_failed` sentinel honoured by observe	`7953585b`
rstos size guard read the wrong dimension	fixed	`61c8ed8a`
ocp-tool flipped a coastline cell's mask+soil only, leaving an inconsistent surface column	masked nearest-neighbour rebuild of the full column	ocp-tool PR #51

The recurring pattern: the guards were right and the plumbing around them defeated them — a correct refusal followed by an unconditional continue is worse than no guard at all.

7 Falsified claims (kept, struck through)

1. ~~Geostrophically-unbalanced remapped state; initialize changed cells in thermal-wind balance.~~ Failed its own offline gate (discrete ∇p noisier than NN fill); ships opt-in only, off by default.
2. ~~The sub-ice ssh BC violation is the blowup.~~ Real bug, fixed, run still died.
3. ~~The per-leg PISM increment is gentle (≈22 m mean).~~ The mean was diluted by the 98% unchanged domain; the distribution is wildly skewed (max 923 m, 1399 fully-opened nodes).
4. ~~The salt-plume parameterization being off explains the warm cavity.~~ The spp-off cavity is the *coldest* in the CORE3 reference runs; the warm cavity was Bug II.
5. ~~The native ICMGG causes the mid-run crash family.~~ The step-0 swap tests proved the native, coastline-following surface set is *required* on the submesh (62 difference-zone cells cannot initialize otherwise); the mid-run family was Bug IV.
6. ~~Weighted-ist cures the crash family.~~ Necessary hygiene for the o2a blend, not sufficient: the pinned OIFS skin regenerated the instability from bounded inputs.
7. ~~The insulation/snow pile-up is the crash root.~~ It raises the loop gain (and is a real climate bias, §5), but the kW events occur on *thin* ice at 49°N; the clean-ini run crashed identically.

8 Method notes

1. **Cheap controls first.** The cold start (geometry vs state), running the fix, plotting the distribution instead of the mean — each killed a confident claim in minutes.
2. **Measurements from a corrupted run are worse than none** (Bug II poisoned Bug I's diagnosis for a week).
3. **"Same code as the stable reference" localizes nothing by itself** — the reference ran the same FESIM solve for 500 years; the difference was the state and, one layer deeper, a switch default that pinned a skin nobody remembered was pinned.
4. **Instrument both ends of a coupling before theorizing about either.** Receive-side logging (ICEDBG) proved the flux was corrupt; send-side decomposition (SENDDBG) named the term and the tile state in its first twelve lines. The two probes cost one rebuild each.
5. **Beware silent fallbacks** (Bug V's `isfile()` miss; the `-ocean given` silent fallback to `th`). A fallback on a nonexistent path should say so.

9 Assets

- Send/receive flux instrumentation: `SENDDBG` (OIFS `A_Q_ice` decomposition + tile state), `ICEDBG` (FESIM received flux + state at cold nodes), `CPLDBG` (OIFS ingest bounds) — all log-only, strip before merging.
- `oasis_expout_o2a` runscript flag: every o2a exchange dumped to netcdf (~20× slowdown; crash-window probes only).
- Clean chunk-1 ice initial state (`prep/clean_ice_ini`): reference year-1849 ice fields index-mapped to the submesh; `map_nod.out` makes such remaps exact.
- A 2-minute offline reproducer (standalone FESOM on the PISM-cavity submesh) and a stable full-mesh control.
- `validate_feom_weights.py`: aborts any leg whose exchange weights are in the wrong node ordering — would have caught Bug II on day one.
- `verify_balance.py`: offline balance/stability gate for remapped restarts.
- Chunk-1 artifacts pooled at `input/fesom2/pism_cavity_ini/` (login-node launches).